

Bridgeless Cuk PFC Converter With Soft Switching in Full Input Voltage and Load Range

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Abstract—In this article, a new bridgeless Cuk power factor correction converter is presented. The proposed converter is capable of providing full soft switching operation for all semiconductor components in the entire line cycle and load variations, only by a single auxiliary switch. The converter operates in discontinuous conduction mode (DCM) that intrinsically provides low input current total harmonic distortion (THD) without any input filter. Despite imposing higher current stress on semiconductors in DCM, the proposed converter achieves high efficiency due to its soft switching operation. The theoretical analysis and operating modes are discussed in detail, and the validity of the analysis is verified by an implemented prototype operating at 200-W output power, 80-V output voltage, and 100-kHz switching frequency, yielding 95.83% efficiency and around 2% THD.

Index Terms—Bridgeless Cuk (BL-Cuk) converter, low total harmonic distortion (THD), power factor correction (PFC), soft switching.

I. INTRODUCTION

NOWADAYS, with the growth of electronic equipment and the utilization of diode bridge rectifiers followed by bulk capacitors as ac–dc converters, harmonic current pollution of the power grid has received more attention. To limit this pollution produced by grid-connected consumers, some standards, such as IEC61000-3-2, are approved [1]. Power supplies must use the power factor correction (PFC) circuit to satisfy these standards. The existence of diode bridge rectifiers in conventional PFCs places too many semiconductor elements in the power flow path, which contributes to higher conduction losses. To eliminate the diode bridge rectifier and enhance the efficiency, bridgeless (BL) PFCs are introduced [2], [3], [4], [5], [6], [7], [8], [9], [10], [11], [12]. To further improve power density and efficiency, passive component size reduction is necessary by increasing the operating frequency.

The soft switching operation is indispensable at high frequency to mitigate switching loss. Soft switching techniques

including Zero-Voltage-Switching (ZVS) and zero-current-switching (ZCS) are applied to the BL structures to diminish the switching loss [12], [13], [14], [15], [16], [17], [18].

The mostly used PFC topology is BL-boost PFC, which has a simple structure, and continuous input current with low total harmonic distortion (THD). But it suffers from high voltage stress, high input inrush current, and the lack of protection against output short circuit. Furthermore, the high output dc level complicates the postregulator stage design [17], [18], [19]. To provide a structure with an output dc level lower than the input voltage, BL-buck PFC is introduced. The existence of the series switch in this topology provides output short-circuit protection and restricts the input inrush current. But the problem of input current zero crossing distortion when the input voltage is lower than the output voltage makes this structure unsuitable for low THD applications [4], [5], [15], [16]. The BL-buck–boost PFC is investigated for eliminating the input current zero crossing distortion problem. Besides, this topology is used for step-up/down voltage gain applications. Due to the series switch of this topology, input inrush current limitation and output short-circuit protection are achieved [20], [21], [22], [23].

Operating in discontinuous conduction mode (DCM) can provide intrinsic PFC with no control circuit complexity. However, extra current stress is imposed on the semiconductors and conduction loss is increased. In addition, due to the discontinuous input current of the PFCs, a large input filter is needed. Continuous conduction mode (CCM) operation reduces the semiconductors current stress but a complex control circuit is required. BL-Cuk and SEPIC PFCs can provide the merits of DCM and CCM operation simultaneously. Consequently, a large input filter, complex control circuit, and high current stress on semiconductors are decreased. In addition, BL-Cuk and SEPIC PFCs can provide output voltage lower or higher than the input voltage with no pulsating input current that provides low input current THD, inherently [6], [7], [8], [9], [10], [11], [12].

A BL-SEPIC converter with no circulating loss is investigated in [7], but this PFC operates at hard switching conditions and CCM mode causing high switching loss and complex control circuit, respectively. Also, the size and the number of magnetic cores restrict this PFC application. In [11], a BL-Cuk PFC is introduced that reduces the number of semiconductors placed in the power flow path. But, it still suffers from hard switching problem. A novel fully soft-switched BL-SEPIC converter is presented in [12], but it employs a high number of semiconductor components and large size inductors along with a complex

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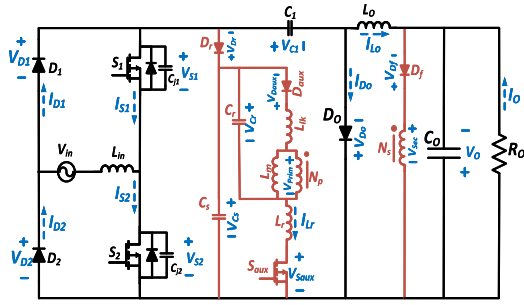


Fig. 1. Proposed converter.

control circuit due to the CCM operation. The BL-buck PFC presented in [15] provides full soft switching and almost eliminates zero-crossing distortion only by a single auxiliary switch. However, it suffers from high voltage and current stress due to its quasi-resonant operation, and the auxiliary switch loses soft switching operation near zero crossing points. In addition, decreasing the output voltage augments the input current THD. A novel quasi-resonant BL-buck PFC converter with almost no zero-crossing distortion is proposed in [16]. The proposed topology provides full soft switching conditions for main components with a simple soft switching cell, but a variable frequency control method is required due to the quasi-resonant operation. Additionally, the auxiliary switch is turned OFF at hard switching conditions. In [20], a new BL-buck-boost PFC with common ground and positive/negative output voltage is proposed, but this PFC suffers from hard switching that makes this topology improper for high-frequency operations.

In this article, a new fully soft switching BL-Cuk PFC converter with intrinsic PFC, limited inrush current, and low input current THD is proposed. The input stage of the proposed converter is similar to the totem-pole topology to reduce the total number of components, while the topology intrinsically provides inrush current protection [3]. The proposed soft switching cell attains soft switching for all semiconductors in the entire input voltage and load variation range. The main and auxiliary switches operation is performed under the ZVS and ZCS conditions, respectively. Soft switching performance along with applying the proposed switching pattern have contributed to improved converter efficiency. Unlike other DCM counterparts, this converter requires no additional large input filter, which improves the power density and efficiency.

II. CIRCUIT DESCRIPTION AND THEORETICAL OPERATION

Fig. 1 shows the proposed Cuk PFC converter, which includes a totem-pole structure and a zero voltage transition (ZVT) cell. The ZVT cell consists of capacitors, C_s and C_r , diodes D_r , D_{aux} , and D_f , inductor L_r , active switch S_{aux} , and a pair of coupled inductors. The ZVT cell provides ZVS conditions for the main switches, S_1 and S_2 , while the auxiliary switch, S_{aux} , is turned ON and OFF under the ZCS condition. According to the ZVT cell structure, the energy of the snubber capacitor, C_s , is recovered to the output. Figs. 2 and 3 illustrate the key waveforms of the proposed PFC converter in one switching cycle and the switching

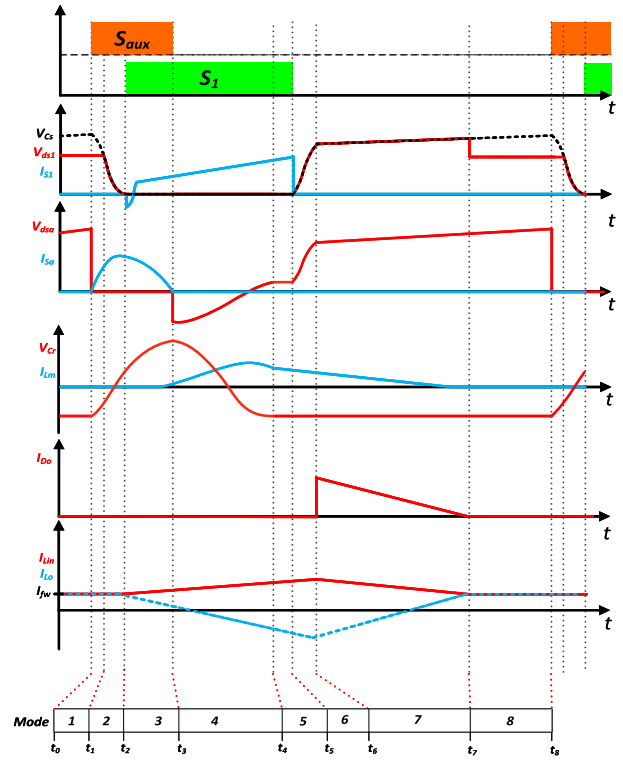


Fig. 2. Key waveforms of the proposed converter.

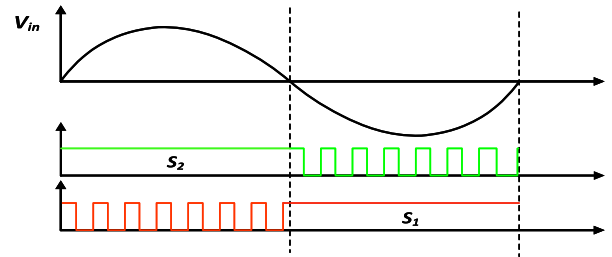


Fig. 3. Switching pattern.

pattern, respectively. To simplify the proposed PFC analysis, some assumptions are made, which include the following:

- 1) the converter operates at steady state;
- 2) all semiconductors are ideal;
- 3) the leakage inductance, L_{lk} is negligible in comparison to the magnetizing inductance, L_m ;
- 4) the output capacitor, C_o , and the input inductor, L_{in} , are large enough that their voltage and current are constant, respectively, in a period;
- 5) switching frequency, f_{sw} , is much higher than the line frequency, f_L , thus, the input voltage, V_{in} , is considered constant in a switching period;
- 6) due to the converter symmetrical operation, only the positive half-line cycle operation is analyzed.

According to the aforementioned assumptions, circuit operation is divided into eight Modes in a switching cycle. Before the first interval, the circuit is in freewheeling mode and the diode D_1 and the body diode of S_2 are conducting and all other

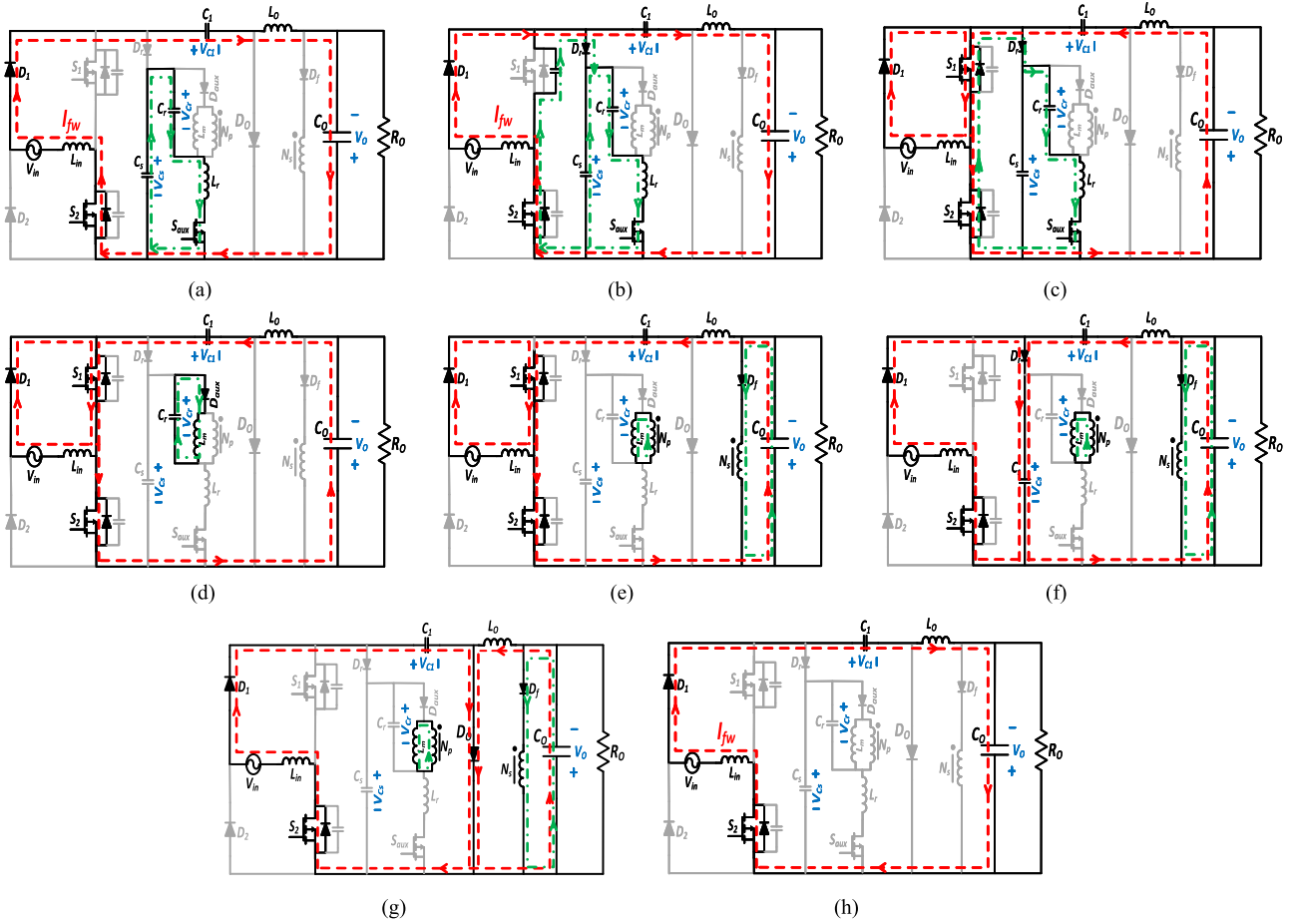


Fig. 4. Equivalent circuit of each mode. (a) Mode 1. (b) Mode 2. (c) Mode 3. (d) Mode 4. (e) Mode 5. (f) Mode 6. (g) Mode 7. (h) Mode 8.

semiconductors are OFF. Also, C_1 and C_s voltages are equal prior to Mode 1.

1) *Mode 1* (t_0 – t_1) [see Fig. 4(a)]: Before turning S_1 ON, S_{aux} is turned ON under ZCS due to the series inductor, L_r , and thus, a resonance occurs among C_s , the resonant capacitor, C_r , and the resonant inductor, L_r . Thereby, the snubber capacitor voltage, V_{C_s} , decreases until it reaches the S_1 drain–source voltage (V_{ds1}). V_{C_s} is equal to the C_1 voltage, V_{C_1} , which is considered constant during a switching cycle and is higher than the input voltage, V_{in} , as described by (1). During this mode, the resonant capacitor voltage, V_{C_r} , increases from its initial value of $-|nV_o|$, where $n = \frac{N_p}{N_s}$ is the coupled inductor turns ratio and V_o is the output voltage. Assuming that the freewheeling current, I_{fw} , which is equal to the input current and is calculated by (2) is constant, the input inductor voltage, $V_{L_{in}}$, and output inductor voltage, V_{L_o} , is zero, and thus, V_{ds1} equals V_{in} . By defining the sum of V_{C_r} and V_{C_s} at t_0 as $V_{C_t}(t_0)$, the current of L_r (I_{L_r}), V_{C_s} , and V_{C_r} are calculated by (4)–(6), respectively. Also The duration of this mode is attained by setting $V_{C_s}(t_1) = V_{ds1}(t_1)$, which yields (7) as follows:

$$V_{C_1}(t) = V_{in}(t) + |V_o| \quad (1)$$

V_{C_s} is equal to V_{C_1} before turning S_{aux} ON. Therefore, during this mode, V_{C_s} is always higher than V_{ds1} .

$$I_{fw} = \frac{2P_o}{\eta V_{in_{peak}}} \quad (2)$$

where η is the converter efficiency, P_o is the output power, and $V_{in_{peak}}$ is the input voltage peak.

$$\begin{aligned} V_{C_t}(t_0) &= |V_{C_s}(t_0)| + |V_{C_r}(t_0)| \\ &= |V_{in}| + |(n+1)V_o|. \end{aligned} \quad (3)$$

$$I_{L_r}(t) = \frac{V_{C_t}(t_0)}{Z_{01}} \sin(\omega_{r1}(t - t_0)) \quad (4)$$

$$\text{where } Z_{01} = \sqrt{\frac{L_r}{C_r C_s}} \text{ and } \omega_{r1} = \left(\sqrt{\frac{L_r C_r C_s}{C_r + C_s}} \right)^{-1}.$$

$$V_{C_s}(t) = \frac{V_{C_t}(t_0)}{C_s \omega_{r1} Z_{01}} \cos(\omega_{r1}(t - t_0)) \quad (5)$$

$$V_{C_r}(t) = -\frac{V_{C_t}(t_0)}{C_r \omega_{r1} Z_{01}} \cos(\omega_{r1}(t - t_0)) \quad (6)$$

$$t_{01} = \frac{1}{\omega_{r1}} \cos^{-1} \left(\frac{V_{in} C_s Z_{01} \omega_{r1}}{V_{C_t}(t_0)} \right). \quad (7)$$

This mode ends when V_{C_s} reaches V_{ds1} .

- 2) *Mode 2* (t_1 – t_2) [see Fig. 4(b)]: At the beginning of this mode, the resonant diode, D_r , turns ON and V_{C_s} continues to decrease until it reaches zero. During this interval, I_{L_r} and V_{C_r} , continue to increase, while V_{C_s} decreases as described by (4)–(6). It is worth noting that S_1 parasitic junction capacitor is much smaller than C_s and can be ignored. Duration of Mode 2 is obtained by $t_{01} + t_{12} = \frac{\pi}{2\omega_{r1}}$. So, t_{12} is calculated by

$$t_{12} = \frac{\pi}{2\omega_{r1}} - \frac{1}{\omega_{r1}} \cos^{-1} \left(\frac{V_{in} C_s Z_{01} \omega_{r1}}{V_{C_t}(t_0)} \right). \quad (8)$$

This mode ends when S_1 is turned ON under the ZVS condition.

- 3) *Mode 3* (t_2 – t_3) [see Fig. 4(c)]: When the body diode of S_1 turns ON, this mode starts and S_1 is turned ON under ZVS. As the result, the input inductor current, $I_{L_{in}}$ starts to increase linearly through S_1 and D_1 ; whereas the output inductor current, I_{L_o} decreases linearly through S_1 and S_2 . A new resonance occurs between L_r and C_r through S_{aux} until I_{L_r} reaches zero, and S_{aux} is turned OFF under the ZCS condition. Because L_m is much higher than L_r , it is assumed that the magnetizing inductance current, I_{L_m} , remains zero. The relations of this interval are defined by (9)–(12) as follows:

$$I_{L_r}(t) = \left(\frac{-V_{C_r}(t_2)}{Z_{02}} \sin(\omega_{r2}(t - t_2)) \right) + \left(\frac{V_{C_t}(t_0)}{Z_{01}} \cos(\omega_{r2}(t - t_2)) \right) \quad (9)$$

where $Z_{02} = \sqrt{\frac{L_r}{C_r}}$ and $\omega_{r2} = \frac{1}{\sqrt{L_r C_r}}$.

$$V_{C_r}(t) = \left(V_{C_r}(t_2) \cos(\omega_{r2}(t - t_2)) \right) + \left(\frac{V_{C_t}(t_0)}{Z_{01}} Z_{02} \sin(\omega_{r2}(t - t_2)) \right) \quad (10)$$

$$I_{L_{in}}(t) = I_{fw} + \frac{V_{in}}{L_{in}}(t - t_2) \quad (11)$$

$$I_{L_o}(t) = I_{fw} - \frac{V_{in}}{L_o}(t - t_2). \quad (12)$$

Duration of this interval can be obtained by

$$t_{23} = \frac{\pi}{2\omega_{r2}}. \quad (13)$$

- 4) *Mode 4* (t_3 – t_4) [see Fig. 4(d)]: At the beginning of this mode, a resonance starts between L_m and C_r . Hence, I_{L_m} increases, while V_{C_r} decreases until it becomes negative, and finally, reaches $-|nV_o|$. Thus

$$I_{L_m}(t) = \frac{V_{C_t}(t_0)}{Z_{01}} Z_{02} \sin(\omega_{r3}(t - t_3)) \quad (14)$$

where $Z_{03} = \sqrt{\frac{L_m}{C_r}}$ and $\omega_{r3} = \frac{1}{\sqrt{L_m C_r}}$.

$$V_{C_r}(t) = |V_{L_m}(t)| = \frac{V_{C_t}(t_0)}{Z_{01}} Z_{02} \cos(\omega_{r3}(t - t_3)). \quad (15)$$

$$t_{34} = \frac{1}{\omega_{r3}} \cos^{-1} \left(\frac{-|nV_o|}{\left| \frac{V_{C_t}(t_0)}{Z_{01}} Z_{02} \right|} \right). \quad (16)$$

At the end of mode3, I_{L_m} is defined by (17) as

$$I_{L_m}(t_{34}) = \frac{\frac{V_{C_t}(t_0)}{Z_{01}} Z_{02}}{Z_{03}} \left[\sqrt{1 - \left(\frac{-|nV_o|}{\left| \frac{V_{C_t}(t_0)}{Z_{01}} Z_{02} \right|} \right)^2} \right] \quad (17)$$

$I_{L_{in}}$ and I_{L_o} continue to increase and decrease, respectively, with the same slope of (11) and (12).

- 5) *Mode 5* (t_4 – t_5) [see Fig. 4(e)]: At the beginning of this mode, the flyback diode, D_f , turns ON and the energy stored in L_m , transfers to the output. Therefore, V_{C_r} remains $-|nV_o|$ and I_{L_m} starts to decrease linearly as follows:

$$I_{L_m}(t) = I_{L_m}(t_{34}) - \frac{|nV_o|}{L_m}(t - t_4). \quad (18)$$

The energy transfer of I_{L_m} continues until it reaches zero in this mode, modes 6 or 7, depending on the input voltage amplitude. Consequently, D_f turns OFF under the ZCS conditions. This mode ends when S_1 is turned OFF under ZVS.

- 6) *Mode 6* (t_5 – t_6) [see Fig. 4(f)]: In this interval, the energy of L_{in} and L_o charge C_s until V_{C_s} reaches V_{C1} and this mode finishes.
- 7) *Mode 7* (t_6 – t_7) [see Fig. 4(g)]: At t_6 , the output diode, D_o , turns ON and the energy stored in L_o transfers to the output. $I_{L_{in}}$ and I_{L_o} start to decrease and increase linearly, respectively, by the equation of

$$I_{L_{in}}(t) = I_{L_{in}}(t_6) - \frac{V_o}{L_{in}}(t - t_6) \quad (19)$$

$$I_{L_o}(t) = I_{L_o}(t_6) + \frac{V_o}{L_o}(t - t_6). \quad (20)$$

When the D_o current reaches zero, it turns OFF under the ZCS condition and this mode ends.

- 8) *Mode 8* (t_7 – t_8) [see Fig. 4(h)]: By turning D_o OFF, the direction of I_{L_o} is reversed and $I_{L_{in}}$ and I_{L_o} become equal and identical to I_{fw} , while V_{ds1} drops to the value of V_{in} . This mode ends by turning the S_{aux} ON and the switching cycle finishes.

III. DESIGN GUIDELINES

A. Design of the Main Converter

The proposed PFC converter is designed for: $|V_o| = 80$ V, $V_{in} = 110V_{rms}$, $f_{sw} = 100$ kHz, $f_L = 50$ Hz, $P_{o_{max}} = 200$ W,

TABLE I
DESIGN PARAMETERS OF THE PROPOSED PFC

Parameters	Value
V_{in}	110 V _{rms}
f_L	50 Hz
f_{sw}	100 kHz
V_o	80 V
L_{in}	1.6 mH
L_o	22 μ H
L_r	10 μ H
L_m	175 μ H
C_1	1 μ F
C_o	2200 μ F
C_s	3.3 nF
C_r	8.2 nF
n	1
S_1, S_2	IRFB4229PBF
S_{aux}	IRFB17N50L+BYV26B
D_1, D_2, D_o	MUR840
D_r, D_{aux}	BYV27-100
D_f	BYV27-600

$I_{ripple} = 0.1I_{in}$. The main converter operates like the conventional BL Cuk converter introduced in [8] and can be designed as reported in [24]. The only difference is the way of calculating D_{sw} . In this article, D_{sw} is calculated by the following relation where 20% dead time is considered to ensure the DCM operation

$$\frac{V_o}{V_{in}} = \frac{D_{sw}}{0.8 - D_{sw}}. \quad (21)$$

The output capacitor is designed like other PFC topologies for voltage ripple around 10% output voltage, which is provided in [12]. The values of the main components are illustrated in Table I.

B. Design of the Auxiliary Circuit

1) Auxiliary Components: The auxiliary components which should be designed include C_s , n , L_r , and C_r . Parameter n influences the auxiliary switch voltage stress ($V_{ds_{aux}}$) according to (3). In addition, to minimize the leakage inductance value, which increases the auxiliary switch voltage stress, $n = 1$ is chosen. Since the main switches soft operation at turn OFF is due to C_s , it can be designed like any other snubber capacitor [25] and is obtained 3.3 nF. Furthermore, C_s , L_r , and C_r affect the peak resonant current ($I_{L_{r_{peak}}}$) and resonant frequency (f_r). To provide soft switching conditions, the auxiliary circuit current must be greater than the freewheeling current in order to forward bias the diode D_r and turn ON the main switch body diode. Thus, the proper value of $I_{L_{r_{peak}}} = 1.5I_{fw}$ is selected and from (2), I_{fw} is calculated around 3 A and $I_{L_{r_{peak}}}$ is obtained 4.5 A. The minimum value of f_r occurs at zero crossing points when C_s is completely depleted and can be calculated by $f_r = \frac{1}{2\pi\sqrt{L_r C_r}}$. To acquire the ZCS condition for the auxiliary switch, $f_r \gg f_{sw}$ must be satisfied, thus $f_r = 5f_{sw}$ is picked. Using (3), (4), and the aforementioned explanations, $L_r = 10 \mu$ H and $C_r = 8.2$ nF are selected. Additionally, to attain S_{aux} ZCS OFF, the following relation must be satisfied:

$$t_{01} + t_{12} + t_{23} \leq D_a T_{sw} \quad (22)$$

where D_a is the auxiliary switch duty cycle and T_{sw} is the switching period. Also, t_{01} , t_{12} , and t_{23} are obtained from (7), (8), and (13), respectively. Thus, $D_a \geq 0.12$ is attained.

2) Designing the L_m : To guarantee DCM operation at all conditions, a dead time should be defined and considered as $0.2T_{sw}$. Thus, charging and discharging of the L_m should occur during $0.8(1 - D_a)T_{sw}$. As the result, according to (23) L_m is obtained about 175 μ H

$$t_{34} + \frac{L_m I_{L_m}(t_{34})}{|nV_o|} = 0.8(1 - D_a)T_{sw}. \quad (23)$$

The design parameters of the proposed PFC are summarized in Table I.

C. Voltage and Current Stress of the Semiconductors

Based on mode6, the voltage stress of switches S_1 and S_2 and diodes D_1 , D_2 , D_f , and D_o are clamped to $V_{in_{max}} + |V_o|$. The voltage stress of S_{aux} equals $V_{in_{max}} + |2V_o|$. The voltage stress of diodes D_r and D_{aux} are clamped to $|V_o|$. The current stress of switch S_{aux} equals the peak of the resonant inductor current. Also as calculated in (11) and (12), the main switches S_1 and S_2 current stress is $(V_{in_{peak}} D_{sw} T_{sw})(L_{in}^{-1} + L_o^{-1})$, where D_{sw} is the main switches duty cycle.

IV. EXPERIMENTAL RESULTS

To verify the performance of the proposed converter and validate the theoretical analysis, a 200-W prototype is implemented. The experimental waveforms and input current THD and PF are provided using Rigol DS1054Z oscilloscope and FLUKE 435-Series II power analyzer, respectively. Fig. 5(a) depicts the input voltage and current, which are almost in phase and the power factor is near unity. The input voltage and current waveforms of the main switch S_1 , the auxiliary switch S_{aux} , and the output diode D_o are shown in Fig. 5(b)–(d), respectively. The output voltage is shown in Fig. 5(e), indicating the voltage ripple is around 10% V_o . Fig. 6(a) and (b) shows the step load change from full load to half load and vice versa, to verify the converter close loop system stability. As observed from Fig. 6(a) and (b), the dynamic time transition from full load to half load and vice versa is around 120 and 110 ms, respectively. Also, the transient response overshoot and undershoot are less than 20% of the output voltage as indicated in the experimental result. Also, Fig. 7(a) and (b) illustrates the efficiency, THD, and PF diagrams versus the output power. Fig. 7(a) compares the proposed converter efficiency with four different types of PFC converters versus the output power. The proposed converter efficiency and THD are approximately 95.8% and 2.1% at full load condition, respectively, which shows a good achievement in the required features. A power loss distribution in detail and the prototype circuit are indicated in Figs. 8 and 9, respectively.

V. COMPARISON

The reported efficiencies of the counterpart converters are presented at various conditions that makes the comparison unfair. In order to have a fair comparison, all the PFC converters

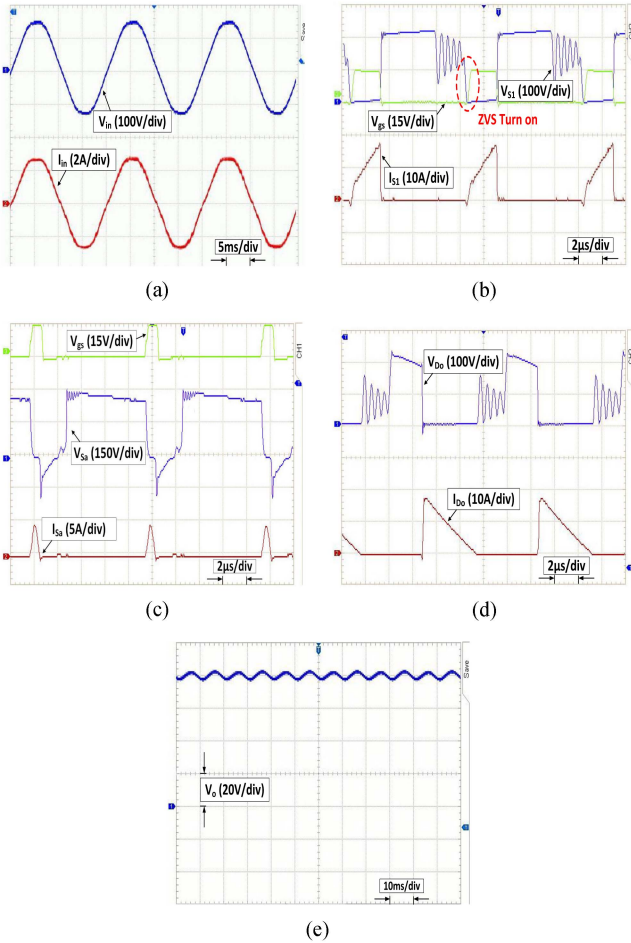


Fig. 5. Experimental waveforms of (a) input voltage and input current; (b) main switch voltage and current; (c) auxiliary switch voltage and current; (d) output diode voltage and current; and (e) output voltage.

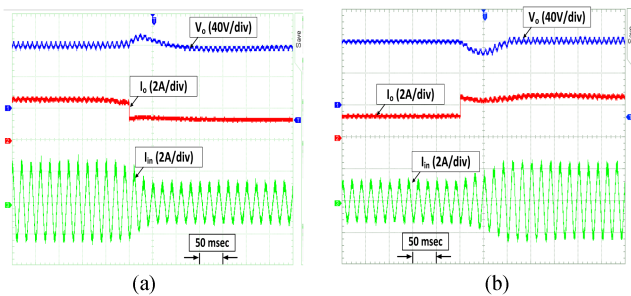
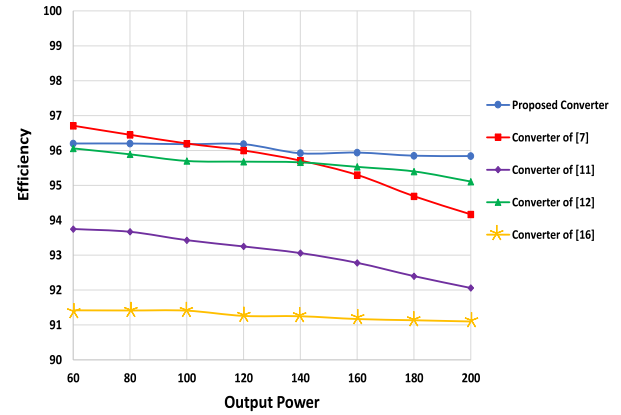
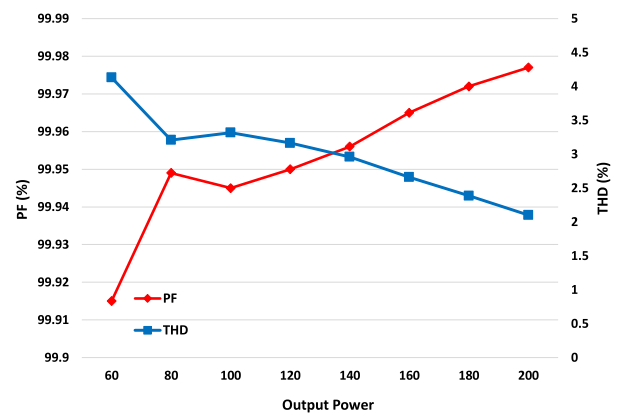


Fig. 6. Experimental waveforms of (a) step load change from full load to half load; and (b) step load change from half load to full load.

are simulated under the same conditions including input voltage, output voltage, and power, switching frequency, and using semiconductors with the same technology. In addition, all circuit parasitic elements are considered in the simulation. Table II presents a comparison among the proposed PFC and other counterpart converters at $P_o = 200$ W, $V_o = 80$ V, $V_{in} = 110V_{rms}$, and $f_{sw} = 100$ kHz. Efficiency, operational mode, THD, output power, soft switching, number of semiconductors, and total elements are considered in this comparison.



(a)



(b)

Fig. 7. (a) Efficiency comparison versus output power. (b) Proposed converter THD and PF versus output power.

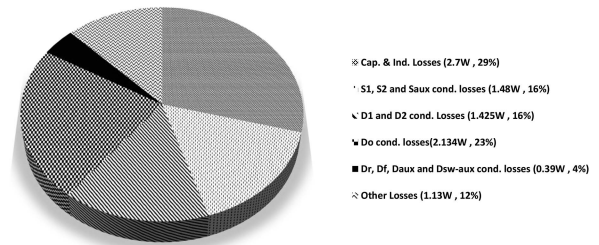


Fig. 8. Power loss distribution at full load condition.

As demonstrated in Table II, the PFCs in [7], [11], and [20] suffer from hard switching that increases switching loss. Despite that the PFCs proposed in [15] and [16] are fully soft switched except for the auxiliary switch, their quasi-resonant operation complicates the control circuit and increases the current stress, which results in efficiency degradation. In addition, the problem of zero crossing distortion in [15] is still not solved. Since, the PFCs in [7], [12], and [20] operate in CCM mode, a complex control circuit is required. Besides, the PFC in [12] has a high component count. By utilizing the soft switching cell in the proposed circuit along with the proposed switching pattern, the

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